

Figure 4.11: (a)
Bisexual flower

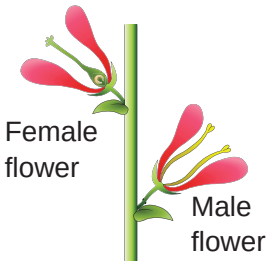


Figure 4.12: (a)
Monoecious

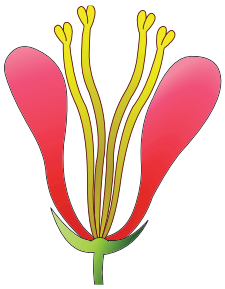


Figure 4.11: (b)
Male flower



Figure 4.12: (b)
Dioecious

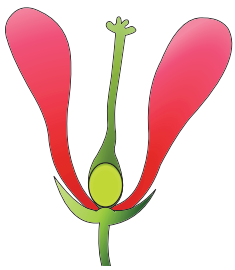


Figure 4.11: (c)
Female flower

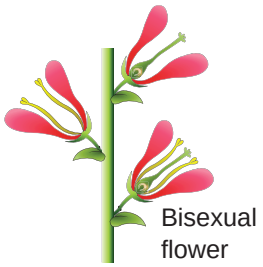


Figure 4.12: (c)
Polygamous

ii) Pistillate flowers: Flowers with only gynoecium are called **pistillate flowers**.

4.2.3 Plant sex

Plant sex refers to the presence and distribution of flowers with different sexes in an individual plant.

1. **Hermaphroditic**: All the flowers of the plant are bisexual.

2. **Monoecious** (*mono*-**one;** *oikos*-**house**): Both male and female flowers are present in the same plant Example: Coconut.

3. **Dioecious** (*di*-**two:** *oikos*-**house**): Male and Female flowers are present on separate plants. Example: Papaya, Palmyra.

Types of Polygamous:

Andromonoecious: A plant with both male flowers and bisexual flowers.

Gynomonoecious: A plant with both pistillate and bisexual flowers.

Polygamomonoecious: A plant with pistillate, staminate and bisexual flowers. It is also called trimonoecious.

Androdioecious: A plant with staminate flowers on one individual and bisexual flower on other individual

Gynodioecious: A plant with pistillate flowers on one individual and bisexual flowers on other individual.

Polygamodioecious: A plant with staminate flowers and bisexual in one individual and pistillate flowers and bisexual flowers in other individual.

Trioecious: A plant with staminate,pistillate and bisexual flowers on different individuals

4. **Polygamous**: The condition in which bisexual and unisexual (staminate/ pistillate) flowers occur in a same plant is called **polygamous**. It is of several types. See box. Example: Musa, *Mangifera*.

4.2.4 Flower symmetry

What is the radius of a circle? Cut a paper into round shape, fold it so as to get two equal halves. In how many planes will you get equal halves? In how many planes you can divide a cucumber in two equal halves? A flower is symmetrical when it is divided into equal halves in any plane running through the center. Flower symmetry